

## 10.2 Graph Terminology and Special Types of Graphs

### 10.2.1 Introduction

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We introduce some of the basic vocabulary of graph theory in this section. We will use this vocabulary later in this chapter when we solve many different types of problems. One such problem involves determining whether a graph can be drawn in the plane so that no two of its edges cross. Another example is deciding whether there is a one-to-one correspondence between the vertices of two graphs that produces a one-to-one correspondence between the edges of the graphs. We will also introduce several important families of graphs often used as examples and in models. Several important applications will be described where these special types of graphs arise.

### 10.2.2 Basic Terminology

First, we give some terminology that describes the vertices and edges of undirected graphs.

#### Definition 1

Two vertices  $u$  and  $v$  in an undirected graph  $G$  are called *adjacent* (or *neighbors*) in  $G$  if  $u$  and  $v$  are endpoints of an edge  $e$  of  $G$ . Such an edge  $e$  is called *incident with* the vertices  $u$  and  $v$  and  $e$  is said to *connect*  $u$  and  $v$ .

We will also find useful terminology describing the set of vertices adjacent to a particular vertex of a graph.

#### Definition 2

The set of all neighbors of a vertex  $v$  of  $G = (V, E)$ , denoted by  $N(v)$ , is called the *neighborhood* of  $v$ . If  $A$  is a subset of  $V$ , we denote by  $N(A)$  the set of all vertices in  $G$  that are adjacent to at least one vertex in  $A$ . So,  $N(A) = \bigcup_{v \in A} N(v)$ .

To keep track of how many edges are incident to a vertex, we make the following definition.

#### Definition 3

The *degree of a vertex in an undirected graph* is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex. The degree of the vertex  $v$  is denoted by  $\deg(v)$ .

#### EXAMPLE 1

What are the degrees and what are the neighborhoods of the vertices in the graphs  $G$  and  $H$  displayed in Figure 1?

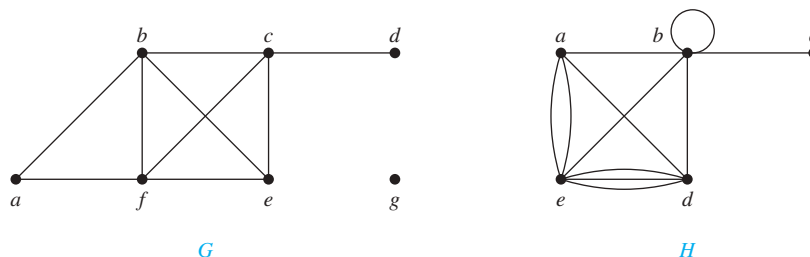


FIGURE 1 The undirected graphs  $G$  and  $H$ .

**Solution:** In  $G$ ,  $\deg(a) = 2$ ,  $\deg(b) = \deg(c) = \deg(f) = 4$ ,  $\deg(d) = 1$ ,  $\deg(e) = 3$ , and  $\deg(g) = 0$ . The neighborhoods of these vertices are  $N(a) = \{b, f\}$ ,  $N(b) = \{a, c, e, f\}$ ,  $N(c) = \{b, d, e, f\}$ ,  $N(d) = \{c\}$ ,  $N(e) = \{b, c, f\}$ ,  $N(f) = \{a, b, c, e\}$ , and  $N(g) = \emptyset$ . In  $H$ ,  $\deg(a) = 4$ ,  $\deg(b) = \deg(e) = 6$ ,  $\deg(c) = 1$ , and  $\deg(d) = 5$ . The neighborhoods of these vertices are  $N(a) = \{b, d, e\}$ ,  $N(b) = \{a, b, c, d, e\}$ ,  $N(c) = \{b\}$ ,  $N(d) = \{a, b, e\}$ , and  $N(e) = \{a, b, d\}$ . ◀

A vertex of degree zero is called **isolated**. It follows that an isolated vertex is not adjacent to any vertex. Vertex  $g$  in graph  $G$  in Example 1 is isolated. A vertex is **pendant** if and only if it has degree one. Consequently, a pendant vertex is adjacent to exactly one other vertex. Vertex  $d$  in graph  $G$  in Example 1 is pendant.

Examining the degrees of vertices in a graph model can provide useful information about the model, as Example 2 shows.

**EXAMPLE 2** What does the degree of a vertex in a niche overlap graph (introduced in Example 11 in Section 10.1) represent? Which vertices in this graph are pendant and which are isolated? Use the niche overlap graph shown in Figure 11 of Section 10.1 to interpret your answers.

**Solution:** There is an edge between two vertices in a niche overlap graph if and only if the two species represented by these vertices compete. Hence, the degree of a vertex in a niche overlap graph is the number of species in the ecosystem that compete with the species represented by this vertex. A vertex is pendant if the species competes with exactly one other species in the ecosystem. Finally, the vertex representing a species is isolated if this species does not compete with any other species in the ecosystem.

For instance, the degree of the vertex representing the squirrel in the niche overlap graph in Figure 11 in Section 10.1 is four, because the squirrel competes with four other species: the crow, the opossum, the raccoon, and the woodpecker. In this niche overlap graph, the mouse is the only species represented by a pendant vertex, because the mouse competes only with the shrew and all other species compete with at least two other species. There are no isolated vertices in the graph in this niche overlap graph because every species in this ecosystem competes with at least one other species. ◀

What do we get when we add the degrees of all the vertices of a graph  $G = (V, E)$ ? Each edge contributes two to the sum of the degrees of the vertices because an edge is incident with exactly two (possibly equal) vertices. This means that the sum of the degrees of the vertices is twice the number of edges. We have the result in Theorem 1, which is sometimes called the handshaking theorem (and is also often known as the handshaking lemma), because of the analogy between an edge having two endpoints and a handshake involving two hands. (Exercise 6 is based on this analogy.)

**THEOREM 1 THE HANDSHAKING THEOREM** Let  $G = (V, E)$  be an undirected graph with  $m$  edges. Then

$$2m = \sum_{v \in V} \deg(v).$$

(Note that this applies even if multiple edges and loops are present.)

**EXAMPLE 3** How many edges are there in a graph with 10 vertices each of degree six?

**Solution:** Because the sum of the degrees of the vertices is  $6 \cdot 10 = 60$ , it follows that  $2m = 60$  where  $m$  is the number of edges. Therefore,  $m = 30$ . ◀

Theorem 1 shows that the sum of the degrees of the vertices of an undirected graph is even. This simple fact has many consequences, one of which is given as Theorem 2.

### THEOREM 2

An undirected graph has an even number of vertices of odd degree.

**Proof:** Let  $V_1$  and  $V_2$  be the set of vertices of even degree and the set of vertices of odd degree, respectively, in an undirected graph  $G = (V, E)$  with  $m$  edges. Then

$$2m = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v).$$

Because  $\deg(v)$  is even for  $v \in V_1$ , the first term in the right-hand side of the last equality is even. Furthermore, the sum of the two terms on the right-hand side of the last equality is even, because this sum is  $2m$ . Hence, the second term in the sum is also even. Because all the terms in this sum are odd, there must be an even number of such terms. Thus, there are an even number of vertices of odd degree.  $\triangleleft$

Terminology for graphs with directed edges reflects the fact that edges in directed graphs have directions.

### Definition 4

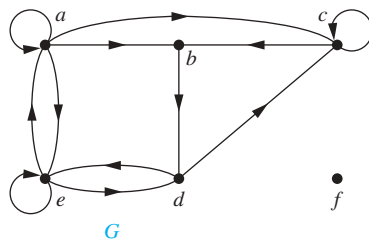
When  $(u, v)$  is an edge of the graph  $G$  with directed edges,  $u$  is said to be *adjacent to*  $v$  and  $v$  is said to be *adjacent from*  $u$ . The vertex  $u$  is called the *initial vertex* of  $(u, v)$ , and  $v$  is called the *terminal* or *end vertex* of  $(u, v)$ . The initial vertex and terminal vertex of a loop are the same.

Because the edges in graphs with directed edges are ordered pairs, the definition of the degree of a vertex can be refined to reflect the number of edges with this vertex as the initial vertex and as the terminal vertex.

### Definition 5

In a graph with directed edges the *in-degree* of a vertex  $v$ , denoted by  $\deg^-(v)$ , is the number of edges with  $v$  as their terminal vertex. The *out-degree* of  $v$ , denoted by  $\deg^+(v)$ , is the number of edges with  $v$  as their initial vertex. (Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.)

**EXAMPLE 4** Find the in-degree and out-degree of each vertex in the graph  $G$  with directed edges shown in Figure 2.



**FIGURE 2** The directed graph  $G$ .

**Solution:** The in-degrees in  $G$  are  $\deg^-(a) = 2$ ,  $\deg^-(b) = 2$ ,  $\deg^-(c) = 3$ ,  $\deg^-(d) = 2$ ,  $\deg^-(e) = 3$ , and  $\deg^-(f) = 0$ . The out-degrees are  $\deg^+(a) = 4$ ,  $\deg^+(b) = 1$ ,  $\deg^+(c) = 2$ ,  $\deg^+(d) = 2$ ,  $\deg^+(e) = 3$ , and  $\deg^+(f) = 0$ . ▶

Because each edge has an initial vertex and a terminal vertex, the sum of the in-degrees and the sum of the out-degrees of all vertices in a graph with directed edges are the same. Both of these sums are the number of edges in the graph. This result is stated as Theorem 3.

### THEOREM 3

Let  $G = (V, E)$  be a graph with directed edges. Then

$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|.$$

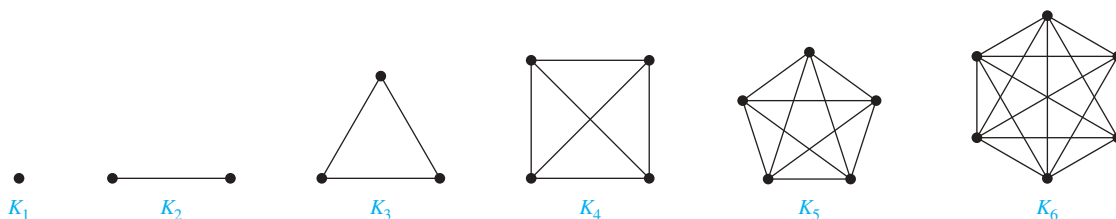
There are many properties of a graph with directed edges that do not depend on the direction of its edges. Consequently, it is often useful to ignore these directions. The undirected graph that results from ignoring directions of edges is called the **underlying undirected graph**. A graph with directed edges and its underlying undirected graph have the same number of edges.

## 10.2.3 Some Special Simple Graphs

We will now introduce several classes of simple graphs. These graphs are often used as examples and arise in many applications.

### EXAMPLE 5

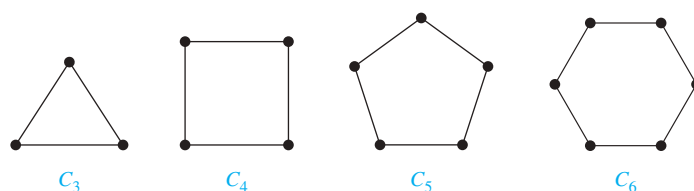
**Complete Graphs** A **complete graph on  $n$  vertices**, denoted by  $K_n$ , is a simple graph that contains exactly one edge between each pair of distinct vertices. The graphs  $K_n$ , for  $n = 1, 2, 3, 4, 5, 6$ , are displayed in Figure 3. A simple graph for which there is at least one pair of distinct vertex not connected by an edge is called **noncomplete**. ▶



**FIGURE 3** The graphs  $K_n$  for  $1 \leq n \leq 6$ .

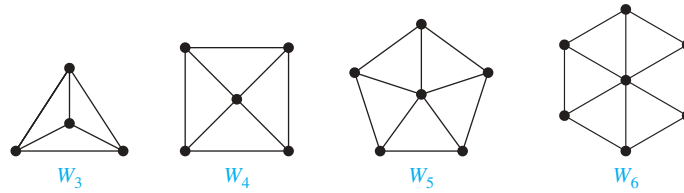
### EXAMPLE 6

**Cycles** A **cycle  $C_n$** ,  $n \geq 3$ , consists of  $n$  vertices  $v_1, v_2, \dots, v_n$  and edges  $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$ , and  $\{v_n, v_1\}$ . The cycles  $C_3, C_4, C_5$ , and  $C_6$  are displayed in Figure 4. ▶



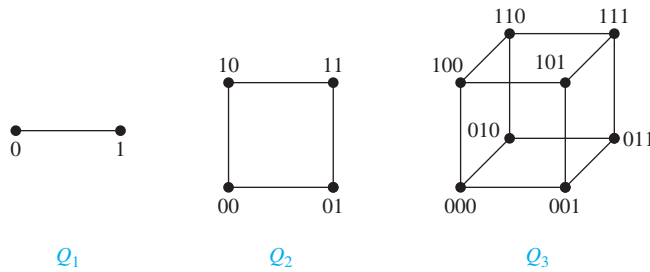
**FIGURE 4** The cycles  $C_3, C_4, C_5$ , and  $C_6$ .

**EXAMPLE 7** **Wheels** We obtain a **wheel**  $W_n$  when we add an additional vertex to a cycle  $C_n$ , for  $n \geq 3$ , and connect this new vertex to each of the  $n$  vertices in  $C_n$ , by new edges. The wheels  $W_3$ ,  $W_4$ ,  $W_5$ , and  $W_6$  are displayed in Figure 5. ◀



**FIGURE 5** The wheels  $W_3$ ,  $W_4$ ,  $W_5$ , and  $W_6$ .

**EXAMPLE 8**  **$n$ -Cubes** An  $n$ -dimensional hypercube, or  $n$ -cube, denoted by  $Q_n$ , is a graph that has vertices representing the  $2^n$  bit strings of length  $n$ . Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position. We display  $Q_1$ ,  $Q_2$ , and  $Q_3$  in Figure 6.



**FIGURE 6** The  $n$ -cube  $Q_n$ ,  $n = 1, 2, 3$ .

Note that you can construct the  $(n + 1)$ -cube  $Q_{n+1}$  from the  $n$ -cube  $Q_n$  by making two copies of  $Q_n$ , prefacing the labels on the vertices with a 0 in one copy of  $Q_n$  and with a 1 in the other copy of  $Q_n$ , and adding edges connecting two vertices that have labels differing only in the first bit. In Figure 6,  $Q_3$  is constructed from  $Q_2$  by drawing two copies of  $Q_2$  as the top and bottom faces of  $Q_3$ , adding 0 at the beginning of the label of each vertex in the bottom face and 1 at the beginning of the label of each vertex in the top face. (Here, by *face* we mean a face of a cube in three-dimensional space. Think of drawing the graph  $Q_3$  in three-dimensional space with copies of  $Q_2$  as the top and bottom faces of a cube and then drawing the projection of the resulting depiction in the plane.) ◀

## 10.2.4 Bipartite Graphs



Sometimes a graph has the property that its vertex set can be divided into two disjoint subsets such that each edge connects a vertex in one of these subsets to a vertex in the other subset. For example, consider the graph representing marriages between men and women in a village, where each person is represented by a vertex and a marriage is represented by an edge. In this graph, each edge connects a vertex in the subset of vertices representing males and a vertex in the subset of vertices representing females. This leads us to Definition 5.

### Definition 6

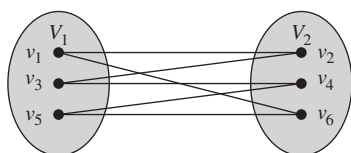
A simple graph  $G$  is called *bipartite* if its vertex set  $V$  can be partitioned into two disjoint sets  $V_1$  and  $V_2$  such that every edge in the graph connects a vertex in  $V_1$  and a vertex in  $V_2$  (so that no edge in  $G$  connects either two vertices in  $V_1$  or two vertices in  $V_2$ ). When this condition holds, we call the pair  $(V_1, V_2)$  a *bipartition* of the vertex set  $V$  of  $G$ .

In Example 9 we will show that  $C_6$  is bipartite, and in Example 10 we will show that  $K_3$  is not bipartite.

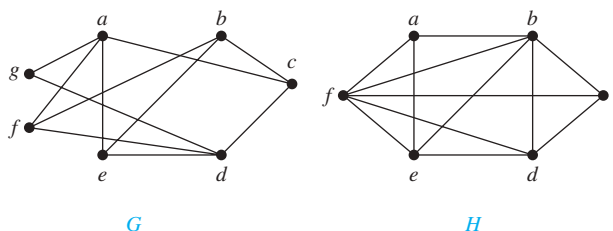
**EXAMPLE 9**  $C_6$  is bipartite, as shown in Figure 7, because its vertex set can be partitioned into the two sets  $V_1 = \{v_1, v_3, v_5\}$  and  $V_2 = \{v_2, v_4, v_6\}$ , and every edge of  $C_6$  connects a vertex in  $V_1$  and a vertex in  $V_2$ .

**EXAMPLE 10**  $K_3$  is not bipartite. To verify this, note that if we divide the vertex set of  $K_3$  into two disjoint sets, one of the two sets must contain two vertices. If the graph were bipartite, these two vertices could not be connected by an edge, but in  $K_3$  each vertex is connected to every other vertex by an edge.

**EXAMPLE 11** Are the graphs  $G$  and  $H$  displayed in Figure 8 bipartite?



**FIGURE 7** Showing that  $C_6$  is bipartite.



**FIGURE 8** The undirected graphs  $G$  and  $H$ .

**Solution:** Graph  $G$  is bipartite because its vertex set is the union of two disjoint sets,  $\{a, b, d\}$  and  $\{c, e, f, g\}$ , and each edge connects a vertex in one of these subsets to a vertex in the other subset. (Note that for  $G$  to be bipartite it is not necessary that every vertex in  $\{a, b, d\}$  be adjacent to every vertex in  $\{c, e, f, g\}$ . For instance,  $b$  and  $g$  are not adjacent.)

Graph  $H$  is not bipartite because its vertex set cannot be partitioned into two subsets so that edges do not connect two vertices from the same subset. (The reader should verify this by considering the vertices  $a, b$ , and  $f$ .)

Theorem 4 provides a useful criterion for determining whether a graph is bipartite.

**THEOREM 4** A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.

**Proof:** First, suppose that  $G = (V, E)$  is a bipartite simple graph. Then  $V = V_1 \cup V_2$ , where  $V_1$  and  $V_2$  are disjoint sets and every edge in  $E$  connects a vertex in  $V_1$  and a vertex in  $V_2$ . If we assign one color to each vertex in  $V_1$  and a second color to each vertex in  $V_2$ , then no two adjacent vertices are assigned the same color.

Now suppose that it is possible to assign colors to the vertices of the graph using just two colors so that no two adjacent vertices are assigned the same color. Let  $V_1$  be the set of vertices assigned one color and  $V_2$  be the set of vertices assigned the other color. Then,  $V_1$  and  $V_2$  are disjoint and  $V = V_1 \cup V_2$ . Furthermore, every edge connects a vertex in  $V_1$  and a vertex in  $V_2$  because no two adjacent vertices are either both in  $V_1$  or both in  $V_2$ . Consequently,  $G$  is bipartite.

We illustrate how Theorem 4 can be used to determine whether a graph is bipartite in Example 12.

**EXAMPLE 12** Use Theorem 4 to determine whether the graphs in Example 11 are bipartite.

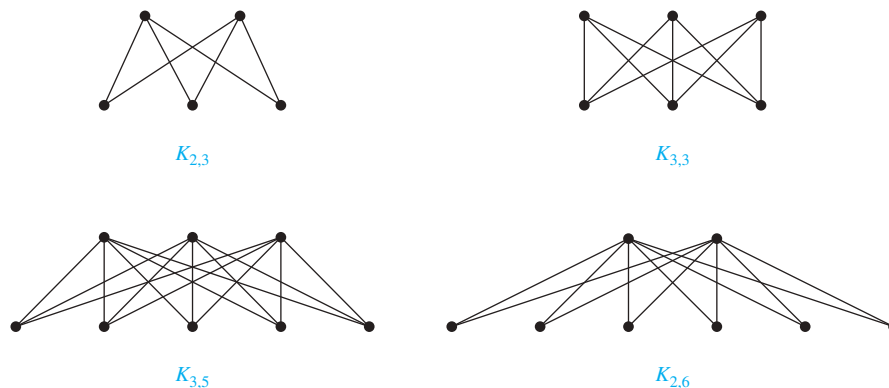
**Solution:** We first consider the graph  $G$ . We will try to assign one of two colors, say red and blue, to each vertex in  $G$  so that no edge in  $G$  connects a red vertex and a blue vertex. Without loss of generality we begin by arbitrarily assigning red to  $a$ . Then, we must assign blue to  $c$ ,  $e$ ,  $f$ , and  $g$ , because each of these vertices is adjacent to  $a$ . To avoid having an edge with two blue endpoints, we must assign red to all the vertices adjacent to either  $c$ ,  $e$ ,  $f$ , or  $g$ . This means that we must assign red to both  $b$  and  $d$  (and means that  $a$  must be assigned red, which it already has been). We have now assigned colors to all vertices, with  $a$ ,  $b$ , and  $d$  red and  $c$ ,  $e$ ,  $f$ , and  $g$  blue. Checking all edges, we see that every edge connects a red vertex and a blue vertex. Hence, by Theorem 4 the graph  $G$  is bipartite.

Next, we will try to assign either red or blue to each vertex in  $H$  so that no edge in  $H$  connects a red vertex and a blue vertex. Without loss of generality we arbitrarily assign red to  $a$ . Then, we must assign blue to  $b$ ,  $e$ , and  $f$ , because each is adjacent to  $a$ . But this is not possible because  $e$  and  $f$  are adjacent, so both cannot be assigned blue. This argument shows that we cannot assign one of two colors to each of the vertices of  $H$  so that no adjacent vertices are assigned the same color. It follows by Theorem 4 that  $H$  is not bipartite. ◀

Theorem 4 is an example of a result in the part of graph theory known as graph colorings. Graph colorings is an important part of graph theory with important applications. We will study graph colorings further in Section 10.8.

Another useful criterion for determining whether a graph is bipartite is based on the notion of a path, a topic we study in Section 10.4. A graph is bipartite if and only if it is not possible to start at a vertex and return to this vertex by traversing an odd number of distinct edges. We will make this notion more precise when we discuss paths and circuits in graphs in Section 10.4 (see Exercise 63 in that section).

**EXAMPLE 13 Complete Bipartite Graphs** A **complete bipartite graph**  $K_{m,n}$  is a graph that has its vertex set partitioned into two subsets of  $m$  and  $n$  vertices, respectively with an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset. The complete bipartite graphs  $K_{2,3}$ ,  $K_{3,3}$ ,  $K_{3,5}$ , and  $K_{2,6}$  are displayed in Figure 9. ◀



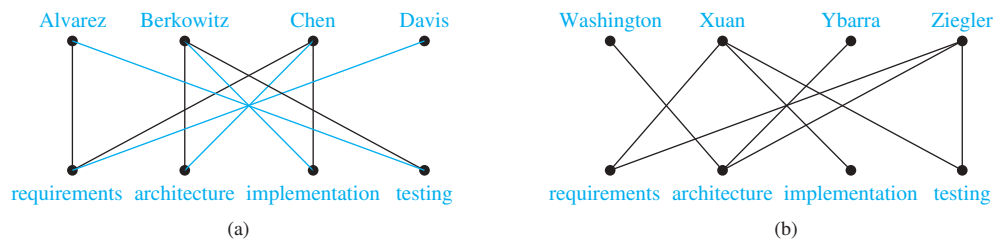
**FIGURE 9** Some complete bipartite graphs.

## 10.2.5 Bipartite Graphs and Matchings

Bipartite graphs can be used to model many types of applications that involve matching the elements of one set to elements of another, as Example 14 illustrates.

**EXAMPLE 14 Job Assignments** Suppose that there are  $m$  employees in a group and  $n$  different jobs that need to be done, where  $m \geq n$ . Each employee is trained to do one or more of these  $n$  jobs. We would like to assign an employee to each job. To help with this task, we can use a graph to model employee capabilities. We represent each employee by a vertex and each job by a vertex. For each employee, we include an edge from that employee to all jobs that the employee has been trained to do. Note that the vertex set of this graph can be partitioned into two disjoint sets, the set of employees and the set of jobs, and each edge connects an employee to a job. Consequently, this graph is bipartite, where the bipartition is  $(E, J)$  where  $E$  is the set of employees and  $J$  is the set of jobs. We now consider two different scenarios.

First, suppose that a group has four employees: Alvarez, Berkowitz, Chen, and Davis; and suppose that four jobs need to be done to complete Project 1: requirements, architecture, implementation, and testing. Suppose that Alvarez has been trained to do requirements and testing; Berkowitz has been trained to do architecture, implementation, and testing; Chen has been trained to do requirements, architecture, and implementation; and Davis has only been trained to do requirements. We model these employee capabilities using the bipartite graph in Figure 10(a).



**FIGURE 10** Modeling the jobs for which employees have been trained.

Second, suppose that a second group also has four employees: Washington, Xuan, Ybarra, and Ziegler; and suppose that the same four jobs need to be done to complete Project 2 as are needed to complete Project 1. Suppose that Washington has been trained to do architecture; Xuan has been trained to do requirements, implementation, and testing; Ybarra has been trained to do architecture; and Ziegler has been trained to do requirements, architecture and testing. We model these employee capabilities using the bipartite graph in Figure 10(b).

To complete Project 1, we must assign an employee to each job so that every job has an employee assigned to it, and so that no employee is assigned more than one job. We can do this by assigning Alvarez to testing, Berkowitz to implementation, Chen to architecture, and Davis to requirements, as shown in Figure 10(a) (where blue lines show this assignment of jobs).

To complete Project 2, we must also assign an employee to each job so that every job has an employee assigned to it and no employee is assigned more than one job. However, this is impossible because there are only two employees, Xuan and Ziegler, who have been trained for at least one of the three jobs of requirements, implementation, and testing. Consequently, there is no way to assign three different employees to these three jobs so that each job is assigned an employee with the appropriate training. ◀

Finding an assignment of jobs to employees can be thought of as finding a matching in the graph model, where a **matching**  $M$  in a simple graph  $G = (V, E)$  is a subset of the set  $E$  of edges of the graph such that no two edges are incident with the same vertex. In other words, a matching is a subset of edges such that if  $\{s, t\}$  and  $\{u, v\}$  are distinct edges of the matching,



then  $s$ ,  $t$ ,  $u$ , and  $v$  are distinct. A vertex that is the endpoint of an edge of a matching  $M$  is said to be **matched** in  $M$ ; otherwise it is said to be **unmatched**. A **maximum matching** is a matching with the largest number of edges. We say that a matching  $M$  in a bipartite graph  $G = (V, E)$  with bipartition  $(V_1, V_2)$  is a **complete matching from  $V_1$  to  $V_2$**  if every vertex in  $V_1$  is the endpoint of an edge in the matching, or equivalently, if  $|M| = |V_1|$ . For example, to assign jobs to employees so that the largest number of jobs are assigned employees, we seek a maximum matching in the graph that models employee capabilities. To assign employees to all jobs we seek a complete matching from the set of jobs to the set of employees. In Example 14, we found a complete matching from the set of jobs to the set of employees for Project 1, and this matching is a maximum matching, and we showed that no complete matching exists from the set of jobs to the employees for Project 2.

We now give an example of how matchings can be used to model marriages.

**EXAMPLE 15 Marriages on an Island** Suppose that there are  $m$  men and  $n$  women on an island. Each person has a list of members of the opposite gender acceptable as a spouse. We construct a bipartite graph  $G = (V_1, V_2)$  where  $V_1$  is the set of men and  $V_2$  is the set of women so that there is an edge between a man and a woman if they find each other acceptable as a spouse. A matching in this graph consists of a set of edges, where each pair of endpoints of an edge is a husband-wife pair. A maximum matching is a largest possible set of married couples, and a complete matching of  $V_1$  is a set of married couples where every man is married, but possibly not all women. ◀

Hall's marriage theorem is an example of a theorem where obvious necessary conditions are sufficient too.

**NECESSARY AND SUFFICIENT CONDITIONS FOR COMPLETE MATCHINGS** We now turn our attention to the question of determining whether a complete matching from  $V_1$  to  $V_2$  exists when  $(V_1, V_2)$  is a bipartition of a bipartite graph  $G = (V, E)$ . We will introduce a theorem that provides a set of necessary and sufficient conditions for the existence of a complete matching. This theorem was proved by Philip Hall in 1935.

**THEOREM 5 HALL'S MARRIAGE THEOREM** The bipartite graph  $G = (V, E)$  with bipartition  $(V_1, V_2)$  has a complete matching from  $V_1$  to  $V_2$  if and only if  $|N(A)| \geq |A|$  for all subsets  $A$  of  $V_1$ .

**Proof:** We first prove the *only if* part of the theorem. To do so, suppose that there is a complete matching  $M$  from  $V_1$  to  $V_2$ . Then, if  $A \subseteq V_1$ , for every vertex  $v \in A$ , there is an edge in  $M$  connecting  $v$  to a vertex in  $V_2$ . Consequently, there are at least as many vertices in  $V_2$  that are neighbors of vertices in  $V_1$  as there are vertices in  $V_1$ . It follows that  $|N(A)| \geq |A|$ .

#### Links



Courtesy of the Edinburgh Mathematical Society

**PHILIP HALL** (1904–1982) Philip Hall grew up in London, where his mother was a dressmaker. He won a scholarship for board school reserved for needy children, and later a scholarship to King's College of Cambridge University. He received his bachelors degree in 1925. In 1926, unsure of his career goals, he took a civil service exam, but decided to continue his studies at Cambridge after failing.

In 1927 Hall was elected to a fellowship at King's College; soon after, he made his first important discovery in group theory. The results he proved are now known as Hall's theorems. In 1933 he was appointed as a Lecturer at Cambridge, where he remained until 1941. During World War II he worked as a cryptographer at Bletchley Park breaking Italian and Japanese codes. At the end of the war, Hall returned to King's College, and was soon promoted. In 1953 he was appointed to the Sadleirian Chair. His work during the 1950s proved to be extremely influential to the rapid development of group theory during the 1960s.

Hall loved poetry and recited it beautifully in Italian and Japanese, as well as English. He was interested in art, music, and botany. He was quite shy and disliked large groups of people. Hall had an incredibly broad and varied knowledge, and was respected for his integrity, intellectual standards, and judgement. He was beloved by his students.



To prove the *if* part of the theorem, the more difficult part, we need to show that if  $|N(A)| \geq |A|$  for all  $A \subseteq V_1$ , then there is a complete matching  $M$  from  $V_1$  to  $V_2$ . We will use strong induction on  $|V_1|$  to prove this.

*Basis step:* If  $|V_1| = 1$ , then  $V_1$  contains a single vertex  $v_0$ . Because  $|N(\{v_0\})| \geq |\{v_0\}| = 1$ , there is at least one edge connecting  $v_0$  and a vertex  $w_0 \in V_2$ . Any such edge forms a complete matching from  $V_1$  to  $V_2$ .

*Inductive step:* We first state the inductive hypothesis.

*Inductive hypothesis:* Let  $k$  be a positive integer. If  $G = (V, E)$  is a bipartite graph with bipartition  $(V_1, V_2)$ , and  $|V_1| = j \leq k$ , then there is a complete matching  $M$  from  $V_1$  to  $V_2$  whenever the condition that  $|N(A)| \geq |A|$  for all  $A \subseteq V_1$  is met.

Now suppose that  $H = (W, F)$  is a bipartite graph with bipartition  $(W_1, W_2)$  and  $|W_1| = k + 1$ . We will prove that the inductive holds using a proof by cases, using two cases. Case (i) applies when for all integers  $j$  with  $1 \leq j \leq k$ , the vertices in every set of  $j$  elements from  $W_1$  are adjacent to at least  $j + 1$  elements of  $W_2$ . Case (ii) applies when for some  $j$  with  $1 \leq j \leq k$  there is a subset  $W'_1$  of  $j$  vertices such that there are exactly  $j$  neighbors of these vertices in  $W_2$ . Because either Case (i) or Case (ii) holds, we need only consider these cases to complete the inductive step.

*Case (i):* Suppose that for all integers  $j$  with  $1 \leq j \leq k$ , the vertices in every subset of  $j$  elements from  $W_1$  are adjacent to at least  $j + 1$  elements of  $W_2$ . Then, we select a vertex  $v \in W_1$  and an element  $w \in N(\{v\})$ , which must exist by our assumption that  $|N(\{v\})| \geq |\{v\}| = 1$ . We delete  $v$  and  $w$  and all edges incident to them from  $H$ . This produces a bipartite graph  $H'$  with bipartition  $(W_1 - \{v\}, W_2 - \{w\})$ . Because  $|W_1 - \{v\}| = k$ , the inductive hypothesis tells us there is a complete matching from  $W_1 - \{v\}$  to  $W_2 - \{w\}$ . Adding the edge from  $v$  to  $w$  to this complete matching produces a complete matching from  $W_1$  to  $W_2$ .

*Case (ii):* Suppose that for some  $j$  with  $1 \leq j \leq k$ , there is a subset  $W'_1$  of  $j$  vertices such that there are exactly  $j$  neighbors of these vertices in  $W_2$ . Let  $W'_2$  be the set of these neighbors. Then, by the inductive hypothesis there is a complete matching from  $W'_1$  to  $W'_2$ . Remove these  $2j$  vertices from  $W_1$  and  $W_2$  and all incident edges to produce a bipartite graph  $K$  with bipartition  $(W_1 - W'_1, W_2 - W'_2)$ .

We will show that the graph  $K$  satisfies the condition  $|N(A)| \geq |A|$  for all subsets  $A$  of  $W_1 - W'_1$ . If not, there would be a subset of  $t$  vertices of  $W_1 - W'_1$  where  $1 \leq t \leq k + 1 - j$  such that the vertices in this subset have fewer than  $t$  vertices of  $W_2 - W'_2$  as neighbors. Then, the set of  $j + t$  vertices of  $W_1$  consisting of these  $t$  vertices together with the  $j$  vertices we removed from  $W_1$  has fewer than  $j + t$  neighbors in  $W_2$ , contradicting the hypothesis that  $|N(A)| \geq |A|$  for all  $A \subseteq W_1$ .

Hence, by the inductive hypothesis, the graph  $K$  has a complete matching. Combining this complete matching with the complete matching from  $W'_1$  to  $W'_2$ , we obtain a complete matching from  $W_1$  to  $W_2$ .

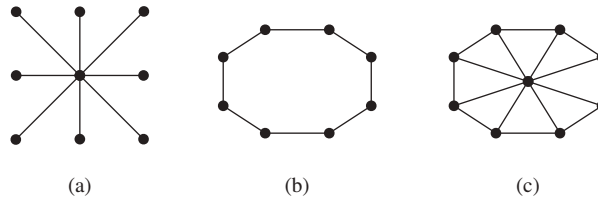
We have shown that in both cases there is a complete matching from  $W_1$  to  $W_2$ . This completes the inductive step and completes the proof. ◀

We have used strong induction to prove Hall's marriage theorem. Although our proof is elegant, it does have some drawbacks. In particular, we cannot construct an algorithm based on this proof that finds a complete matching in a bipartite graph. For a constructive proof that can be used as the basis of an algorithm, see [Gi85].

## 10.2.6 Some Applications of Special Types of Graphs

We conclude this section by introducing some additional graph models that involve the special types of graph we have discussed in this section.

**EXAMPLE 16 Local Area Networks** The various computers in a building, such as minicomputers and personal computers, as well as peripheral devices such as printers and plotters, can be connected using a *local area network*. Some of these networks are based on a *star topology*, where all devices are connected to a central control device. A local area network can be represented using a complete bipartite graph  $K_{1,n}$ , as shown in Figure 11(a). Messages are sent from device to device through the central control device.



**FIGURE 11** Star, ring, and hybrid topologies for local area networks.

Other local area networks are based on a *ring topology*, where each device is connected to exactly two others. Local area networks with a ring topology are modeled using  $n$ -cycles,  $C_n$ , as shown in Figure 11(b). Messages are sent from device to device around the cycle until the intended recipient of a message is reached.

Finally, some local area networks use a hybrid of these two topologies. Messages may be sent around the ring, or through a central device. This redundancy makes the network more reliable. Local area networks with this redundancy can be modeled using wheels  $W_n$ , as shown in Figure 11(c).

**EXAMPLE 17 Interconnection Networks for Parallel Computation** For many years, computers executed programs one operation at a time. Consequently, the algorithms written to solve problems were designed to perform one step at a time; such algorithms are called **serial**. (Almost all algorithms described in this book are serial.) However, many computationally intense problems, such as weather simulations, medical imaging, and cryptanalysis, cannot be solved in a reasonable amount of time using serial operations, even on a supercomputer. Furthermore, there is a physical limit to how fast a computer can carry out basic operations, so there will always be problems that cannot be solved in a reasonable length of time using serial operations.

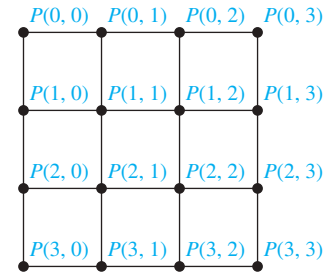
**Parallel processing**, which uses computers made up of many separate processors, each with its own memory, helps overcome the limitations of computers with a single processor. **Parallel algorithms**, which break a problem into a number of subproblems that can be solved concurrently, can then be devised to rapidly solve problems using a computer with multiple processors. In a parallel algorithm, a single instruction stream controls the execution of the algorithm, sending subproblems to different processors, and directs the input and output of these subproblems to the appropriate processors.

When parallel processing is used, one processor may need output generated by another processor. Consequently, these processors need to be interconnected. We can use the appropriate type of graph to represent the interconnection network of the processors in a computer with multiple processors. In the following discussion, we will describe the most commonly used types of interconnection networks for parallel processors. The type of interconnection network used to implement a particular parallel algorithm depends on the requirements for exchange of data between processors, the desired speed, and, of course, the available hardware.

The simplest, but most expensive, network-interconnecting processors include a two-way link between each pair of processors. This network can be represented by  $K_n$ , the complete graph on  $n$  vertices, when there are  $n$  processors. However, there are serious problems with this type of interconnection network because the required number of connections is so large. In reality, the number of direct connections to a processor is limited, so when there are a large number of



**FIGURE 12** A linear array for six processors.



**FIGURE 13** A mesh network for 16 processors.

processors, a processor cannot be linked directly to all others. For example, when there are 64 processors,  $C(64, 2) = 2016$  connections would be required, and each processor would have to be directly connected to 63 others.

On the other hand, perhaps the simplest way to interconnect  $n$  processors is to use an arrangement known as a **linear array**. Each processor  $P_i$ , other than  $P_1$  and  $P_n$ , is connected to its neighbors  $P_{i-1}$  and  $P_{i+1}$  via a two-way link.  $P_1$  is connected only to  $P_2$ , and  $P_n$  is connected only to  $P_{n-1}$ . The linear array for six processors is shown in Figure 12. The advantage of a linear array is that each processor has at most two direct connections to other processors. The disadvantage is that it is sometimes necessary to use a large number of intermediate links, called **hops**, for processors to share information.

The **mesh network** (or **two-dimensional array**) is a commonly used interconnection network. In such a network, the number of processors is a perfect square, say  $n = m^2$ . The  $n$  processors are labeled  $P(i, j)$ ,  $0 \leq i \leq m - 1$ ,  $0 \leq j \leq m - 1$ . Two-way links connect processor  $P(i, j)$  with its four neighbors, processors  $P(i \pm 1, j)$  and  $P(i, j \pm 1)$ , as long as these are processors in the mesh. (Note that four processors, on the corners of the mesh, have only two adjacent processors, and other processors on the boundaries have only three neighbors. Sometimes a variant of a mesh network in which every processor has exactly four connections is used; see Exercise 74.) The mesh network limits the number of links for each processor. Communication between some pairs of processors requires  $O(\sqrt{n}) = O(m)$  intermediate links. (See Exercise 75.) The graph representing the mesh network for 16 processors is shown in Figure 13.

One important type of interconnection network is the hypercube. For such a network, the number of processors is a power of 2,  $n = 2^m$ . The  $n$  processors are labeled  $P_0, P_1, \dots, P_{n-1}$ . Each processor has two-way connections to  $m$  other processors. Processor  $P_i$  is linked to the processors with indices whose binary representations differ from the binary representation of  $i$  in exactly one bit. The hypercube network balances the number of direct connections for each processor and the number of intermediate connections required so that processors can communicate. Many computers have been built using a hypercube network, and many parallel algorithms have been devised that use a hypercube network. The graph  $Q_m$ , the  $m$ -cube, represents the hypercube network with  $n = 2^m$  processors. Figure 14 displays the hypercube network for eight processors. (Figure 14 displays a different way to draw  $Q_3$  than was shown in Figure 6.)

## 10.2.7 New Graphs from Old

Sometimes we need only part of a graph to solve a problem. For instance, we may care only about the part of a large computer network that involves the computer centers in New York, Denver, Detroit, and Atlanta. Then we can ignore the other computer centers and all telephone lines not linking two of these specific four computer centers. In the graph model for the large

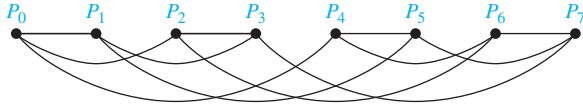
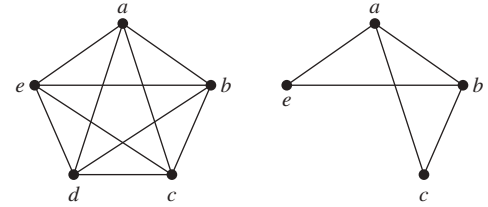


FIGURE 14 A hypercube network for eight processors.

FIGURE 15 A subgraph of  $K_5$ .

network, we can remove the vertices corresponding to the computer centers other than the four of interest, and we can remove all edges incident with a vertex that was removed. When edges and vertices are removed from a graph, without removing endpoints of any remaining edges, a smaller graph is obtained. Such a graph is called a **subgraph** of the original graph.

**Definition 7**

A *subgraph* of a graph  $G = (V, E)$  is a graph  $H = (W, F)$ , where  $W \subseteq V$  and  $F \subseteq E$ . A subgraph  $H$  of  $G$  is a *proper subgraph* of  $G$  if  $H \neq G$ .

Given a set of vertices of a graph, we can form a subgraph of this graph with these vertices and the edges of the graph that connect them.

**Definition 8**

Let  $G = (V, E)$  be a simple graph. The **subgraph induced** by a subset  $W$  of the vertex set  $V$  is the graph  $(W, F)$ , where the edge set  $F$  contains an edge in  $E$  if and only if both endpoints of this edge are in  $W$ .

**EXAMPLE 18** The graph  $G$  shown in Figure 15 is a subgraph of  $K_5$ . If we add the edge connecting  $c$  and  $e$  to  $G$ , we obtain the subgraph induced by  $W = \{a, b, c, e\}$ . ◀

**REMOVING OR ADDING EDGES OF A GRAPH** Given a graph  $G = (V, E)$  and an edge  $e \in E$ , we can produce a subgraph of  $G$  by removing the edge  $e$ . The resulting subgraph, denoted by  $G - e$ , has the same vertex set  $V$  as  $G$ . Its edge set is  $E - \{e\}$ . Hence,

$$G - e = (V, E - \{e\}).$$

Similarly, if  $E'$  is a subset of  $E$ , we can produce a subgraph of  $G$  by removing the edges in  $E'$  from the graph. The resulting subgraph has the same vertex set  $V$  as  $G$ . Its edge set is  $E - E'$ .

We can also add an edge  $e$  to a graph to produce a new larger graph when this edge connects two vertices already in  $G$ . We denote by  $G + e$  the new graph produced by adding a new edge  $e$ , connecting two previously nonincident vertices, to the graph  $G$ . Hence,

$$G + e = (V, E \cup \{e\}).$$

The vertex set of  $G + e$  is the same as the vertex set of  $G$  and the edge set is the union of the edge set of  $G$  and the set  $\{e\}$ . (See Example 19 for examples of removing an edge from a graph and adding an edge to a graph.)

**EDGE CONTRACTIONS** Sometimes when we remove an edge from a graph, we do not want to retain the endpoints of this edge as separate vertices in the resulting subgraph. In such a case we perform an **edge contraction**, which removes an edge  $e$  with endpoints  $u$  and  $v$  and merges  $u$

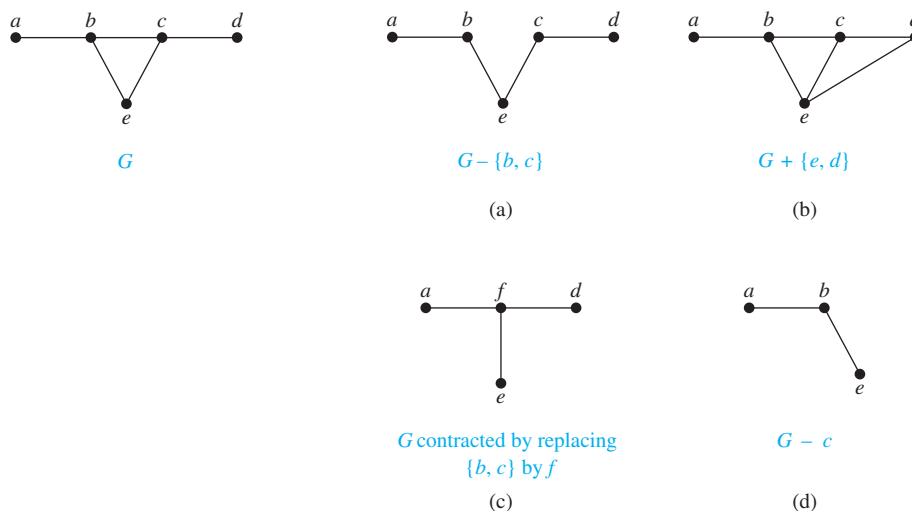
and  $w$  into a new single vertex  $w$ , and for each edge with  $u$  or  $v$  as an endpoint replaces the edge with one with  $w$  as endpoint in place of  $u$  or  $v$  and with the same second endpoint. Hence, the contraction of the edge  $e$  with endpoints  $u$  and  $v$  in the graph  $G = (V, E)$  produces a new graph  $G' = (V', E')$  (which is not a subgraph of  $G$ ), where  $V' = V - \{u, v\} \cup \{w\}$  and  $E'$  contains the edges in  $E$  which do not have either  $u$  or  $v$  as endpoints and an edge connecting  $w$  to every neighbor of either  $u$  or  $v$  in  $V$ . For example, the contraction of the edge connecting the vertices  $e$  and  $c$  in the graph  $G_1$  in Figure 16 produces a new graph  $G'_1$  with vertices  $a, b, d$ , and  $w$ . As in  $G_1$ , there is an edge in  $G'_1$  connecting  $a$  and  $b$  and an edge connecting  $a$  and  $d$ . There also is an edge in  $G'_1$  that connects  $b$  and  $w$  that replaces the edges connecting  $b$  and  $c$  and connecting  $b$  and  $e$  in  $G_1$  and an edge in  $G'_1$  that connects  $d$  and  $w$  replacing the edge connecting  $d$  and  $e$  in  $G_1$ . (Also, see Example 19 for an example of the contraction of an edge in a graph.)

**REMOVING VERTICES FROM A GRAPH** When we remove a vertex  $v$  and all edges incident to it from  $G = (V, E)$ , we produce a subgraph, denoted by  $G - v$ . Observe that  $G - v = (V - \{v\}, E')$ , where  $E'$  is the set of edges of  $G$  not incident to  $v$ . Similarly, if  $V'$  is a subset of  $V$ , then the graph  $G - V'$  is the subgraph  $(V - V', E')$ , where  $E'$  is the set of edges of  $G$  not incident to a vertex in  $V'$ . (See Example 19 for an example of the removal of a vertex from a graph.)

**EXAMPLE 19** Figure 16 displays an undirected graph  $G$  with four different graphs that are the result of different operations on  $G$ . These are:

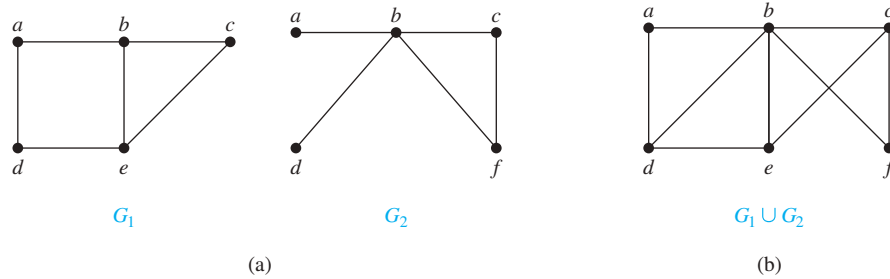
- (a)  $G - \{b, c\}$ , constructed from  $G$  by removing the edge  $\{b, c\}$
- (b)  $G + \{e, d\}$ , constructed from  $G$  by adding the edge  $\{e, d\}$
- (c) the contraction of  $G$ , constructed from  $G$  by replacing the edge  $\{b, c\}$  with a new vertex  $f$ , and replacing the edges  $\{c, d\}$ ,  $\{a, b\}$ ,  $\{b, e\}$ , and  $\{c, e\}$  with the new edges  $\{a, f\}$ ,  $\{f, d\}$ , and  $\{f, e\}$
- (d)  $G - c$ , constructed from  $G$  by removing the vertex  $c$  and the edges  $\{b, c\}$ ,  $\{c, d\}$  and  $\{c, e\}$

**GRAPH UNIONS** Two or more graphs can be combined in various ways. The new graph that contains all the vertices and edges of these graphs is called the **union** of the graphs. We will give a more formal definition for the union of two simple graphs.



**FIGURE 16** The graph  $G$  and four graphs resulting from different operations on  $G$ .





**FIGURE 17** (a) The simple graphs  $G_1$  and  $G_2$ . (b) Their union  $G_1 \cup G_2$ .

### Definition 9

The *union* of two simple graphs  $G_1 = (V_1, E_1)$  and  $G_2 = (V_2, E_2)$  is the simple graph with vertex set  $V_1 \cup V_2$  and edge set  $E_1 \cup E_2$ . The union of  $G_1$  and  $G_2$  is denoted by  $G_1 \cup G_2$ .

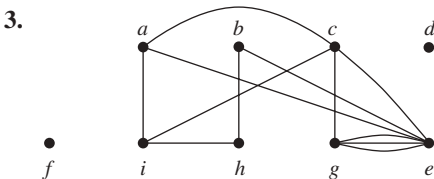
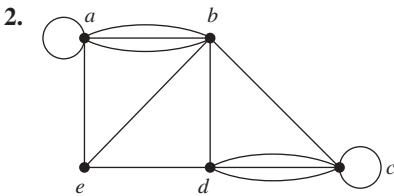
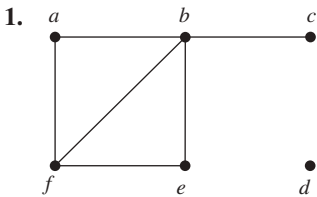
### EXAMPLE 20

Find the union of the graphs  $G_1$  and  $G_2$  shown in Figure 17(a).

**Solution:** The vertex set of the union  $G_1 \cup G_2$  is the union of the two vertex sets, namely,  $\{a, b, c, d, e, f\}$ . The edge set of the union is the union of the two edge sets. The union is displayed in Figure 17(b).

## Exercises

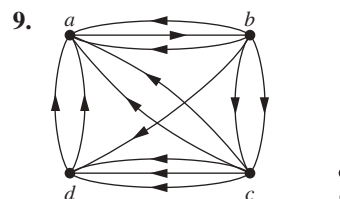
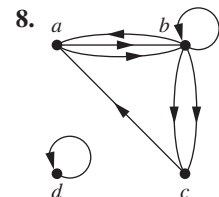
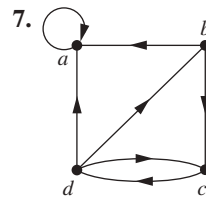
In Exercises 1–3 find the number of vertices, the number of edges, and the degree of each vertex in the given undirected graph. Identify all isolated and pendant vertices.



4. Find the sum of the degrees of the vertices of each graph in Exercises 1–3 and verify that it equals twice the number of edges in the graph.

5. Can a simple graph exist with 15 vertices each of degree five?
6. Show that the sum, over the set of people at a party, of the number of people a person has shaken hands with, is even. Assume that no one shakes his or her own hand.

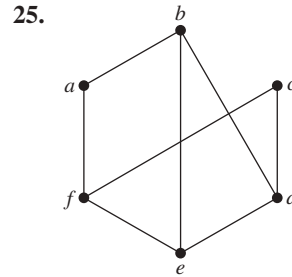
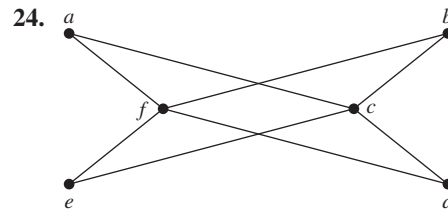
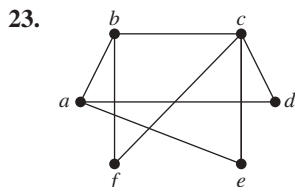
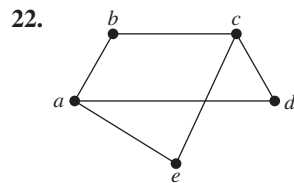
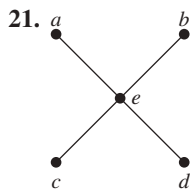
In Exercises 7–9 determine the number of vertices and edges and find the in-degree and out-degree of each vertex for the given directed multigraph.



10. For each of the graphs in Exercises 7–9 determine the sum of the in-degrees of the vertices and the sum of the out-degrees of the vertices directly. Show that they are both equal to the number of edges in the graph.

11. Construct the underlying undirected graph for the graph with directed edges in Figure 2.
12. What does the degree of a vertex represent in the acquaintanceship graph, where vertices represent all the people in the world? What does the neighborhood of a vertex in this graph represent? What do isolated and pendant vertices in this graph represent? In one study it was estimated that the average degree of a vertex in this graph is 1000. What does this mean in terms of the model?
13. What does the degree of a vertex represent in an academic collaboration graph? What does the neighborhood of a vertex represent? What do isolated and pendant vertices represent?
14. What does the degree of a vertex in the Hollywood graph represent? What does the neighborhood of a vertex represent? What do the isolated and pendant vertices represent?
15. What do the in-degree and the out-degree of a vertex in a telephone call graph, as described in Example 4 of Section 10.1, represent? What does the degree of a vertex in the undirected version of this graph represent?
16. What do the in-degree and the out-degree of a vertex in the web graph, as described in Example 5 of Section 10.1, represent?
17. What do the in-degree and the out-degree of a vertex in a directed graph modeling a round-robin tournament represent?
18. Show that in a simple graph with at least two vertices there must be two vertices that have the same degree.
19. Use Exercise 18 to show that in a group of people, there must be two people who are friends with the same number of other people in the group.
20. Draw these graphs.
  - a)  $K_7$                       b)  $K_{1,8}$                       c)  $K_{4,4}$
  - d)  $C_7$                         e)  $W_7$                         f)  $Q_4$

In Exercises 21–25 determine whether the graph is bipartite. You may find it useful to apply Theorem 4 and answer the question by determining whether it is possible to assign either red or blue to each vertex so that no two adjacent vertices are assigned the same color.



26. For which values of  $n$  are these graphs bipartite?
  - a)  $K_n$                       b)  $C_n$                       c)  $W_n$                       d)  $Q_n$
27. Suppose that there are four employees in the computer support group of the School of Engineering of a large university. Each employee will be assigned to support one of four different areas: hardware, software, networking, and wireless. Suppose that Ping is qualified to support hardware, networking, and wireless; Quiggley is qualified to support software and networking; Ruiz is qualified to support networking and wireless, and Sitea is qualified to support hardware and software.
  - a) Use a bipartite graph to model the four employees and their qualifications.
  - b) Use Hall's theorem to determine whether there is an assignment of employees to support areas so that each employee is assigned one area to support.
  - c) If an assignment of employees to support areas so that each employee is assigned to one support area exists, find one.
28. Suppose that a new company has five employees: Zamora, Agraharam, Smith, Chou, and Macintyre. Each employee will assume one of six responsibilities: planning, publicity, sales, marketing, development, and industry relations. Each employee is capable of doing one or more of these jobs: Zamora could do planning, sales, marketing, or industry relations; Agraharam could do planning or development; Smith could do publicity, sales, or industry relations; Chou could do planning, sales, or industry relations; and Macintyre could do planning, publicity, sales, or industry relations.
  - a) Model the capabilities of these employees using a bipartite graph.
  - b) Find an assignment of responsibilities such that each employee is assigned one responsibility.
  - c) Is the matching of responsibilities you found in part (b) a complete matching? Is it a maximum matching?
29. Suppose that there are five young women and five young men on an island. Each man is willing to marry some of the women on the island and each woman is willing to



marry any man who is willing to marry her. Suppose that Sandeep is willing to marry Tina and Vandana; Barry is willing to marry Tina, Xia, and Uma; Teja is willing to marry Tina and Zelda; Anil is willing to marry Vandana and Zelda; and Emilio is willing to marry Tina and Zelda. Use Hall's theorem to show there is no matching of the young men and young women on the island such that each young man is matched with a young woman he is willing to marry.

30. Suppose that there are five young women and six young men on an island. Each woman is willing to marry some of the men on the island and each man is willing to marry any woman who is willing to marry him. Suppose that Anna is willing to marry Jason, Larry, and Matt; Barbara is willing to marry Kevin and Larry; Carol is willing to marry Jason, Nick, and Oscar; Diane is willing to marry Jason, Larry, Nick, and Oscar; and Elizabeth is willing to marry Jason and Matt.
- Model the possible marriages on the island using a bipartite graph.
  - Find a matching of the young women and the young men on the island such that each young woman is matched with a young man whom she is willing to marry.
  - Is the matching you found in part (b) a complete matching? Is it a maximum matching?

Each of Exercises 31–33 can be solved using Hall's theorem.

- \*31. Suppose there is an integer  $k$  such that every man on a desert island is willing to marry exactly  $k$  of the women on the island and every woman on the island is willing to marry exactly  $k$  of the men. Also, suppose that a man is willing to marry a woman if and only if she is willing to marry him. Show that it is possible to match the men and women on the island so that everyone is matched with someone that they are willing to marry.
- \*32. Suppose that  $2n$  tennis players compete in a round-robin tournament. Every player has exactly one match with every other player during  $2n - 1$  consecutive days. Every match has a winner and a loser. Show that it is possible to select a winning player each day without selecting the same player twice.
- \*33. Suppose that  $m$  people are selected as prize winners in a lottery, where each winner can select two prizes from a collection of different prizes. Show if there are  $2m$  prizes that every winner wants, then every winner is able to select two prizes that they want.
- \*34. In this exercise we prove a theorem of Øystein Ore. Suppose that  $G = (V, E)$  is a bipartite graph with bipartition  $(V_1, V_2)$  and that  $A \subseteq V_1$ . Show that the maximum number of vertices of  $V_1$  that are the endpoints of a matching of  $G$  equals  $|V_1| - \max_{A \subseteq V_1} \text{def}(A)$ , where  $\text{def}(A) = |A| - |N(A)|$ . (Here,  $\text{def}(A)$  is called the **deficiency** of  $A$ .) [Hint: Form a larger graph by adding  $\max_{A \subseteq V_1} \text{def}(A)$  new vertices to  $V_2$  and connect all of them to the vertices of  $V_1$ .]

35. For the graph  $G$  in Exercise 1 find
- the subgraph induced by the vertices  $a, b, c$ , and  $f$ .
  - the new graph  $G_1$  obtained from  $G$  by contracting the edge connecting  $b$  and  $f$ .

36. Let  $n$  be a positive integer. Show that a subgraph induced by a nonempty subset of the vertex set of  $K_n$  is a complete graph.
37. How many vertices and how many edges do these graphs have?
- $K_n$
  - $C_n$
  - $W_n$
  - $K_{m,n}$
  - $Q_n$

The **degree sequence** of a graph is the sequence of the degrees of the vertices of the graph in nonincreasing order. For example, the degree sequence of the graph  $G$  in Example 1 is 4, 4, 4, 3, 2, 1, 0.

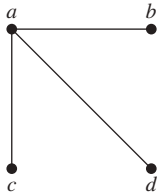
38. Find the degree sequences for each of the graphs in Exercises 21–25.
39. Find the degree sequence of each of the following graphs.
- $K_4$
  - $C_4$
  - $W_4$
  - $K_{2,3}$
  - $Q_3$
40. What is the degree sequence of the bipartite graph  $K_{m,n}$  where  $m$  and  $n$  are positive integers? Explain your answer.
41. What is the degree sequence of  $K_n$ , where  $n$  is a positive integer? Explain your answer.
42. How many edges does a graph have if its degree sequence is 4, 3, 3, 2, 2? Draw such a graph.
43. How many edges does a graph have if its degree sequence is 5, 2, 2, 2, 2, 1? Draw such a graph.

A sequence  $d_1, d_2, \dots, d_n$  is called **graphic** if it is the degree sequence of a simple graph.

44. Determine whether each of these sequences is graphic. For those that are, draw a graph having the given degree sequence.
- 5, 4, 3, 2, 1, 0
  - 6, 5, 4, 3, 2, 1
  - 2, 2, 2, 2, 2, 2
  - 3, 3, 3, 2, 2, 2
  - 3, 3, 2, 2, 2, 2
  - 1, 1, 1, 1, 1, 1
  - 5, 3, 3, 3, 3, 3
  - 5, 5, 4, 3, 2, 1
45. Determine whether each of these sequences is graphic. For those that are, draw a graph having the given degree sequence.
- 3, 3, 3, 3, 2
  - 5, 4, 3, 2, 1
  - 4, 4, 3, 2, 1
  - 4, 4, 3, 3, 3
  - 3, 2, 2, 1, 0
  - 1, 1, 1, 1, 1

- \*46. Suppose that  $d_1, d_2, \dots, d_n$  is a graphic sequence. Show that there is a simple graph with vertices  $v_1, v_2, \dots, v_n$  such that  $\deg(v_i) = d_i$  for  $i = 1, 2, \dots, n$  and  $v_1$  is adjacent to  $v_2, \dots, v_{d_1+1}$ .
- \*47. Show that a sequence  $d_1, d_2, \dots, d_n$  of nonnegative integers in nonincreasing order is a graphic sequence if and only if the sequence obtained by reordering the terms of the sequence  $d_2 - 1, \dots, d_{d_1+1} - 1, d_{d_1+2}, \dots, d_n$  so that the terms are in nonincreasing order is a graphic sequence.
- \*48. Use Exercise 47 to construct a recursive algorithm for determining whether a nonincreasing sequence of positive integers is graphic.

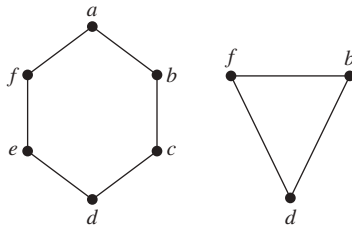
49. Show that every nonincreasing sequence of nonnegative integers with an even sum of its terms is the degree sequence of a pseudograph, that is, an undirected graph where loops are allowed. [Hint: Construct such a graph by first adding as many loops as possible at each vertex. Then add additional edges connecting vertices of odd degree. Explain why this construction works.]
50. How many subgraphs with at least one vertex does  $K_2$  have?
51. How many subgraphs with at least one vertex does  $K_3$  have?
52. How many subgraphs with at least one vertex does  $W_3$  have?
53. Draw all subgraphs of this graph.



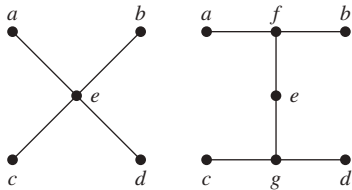
54. Let  $G$  be a graph with  $v$  vertices and  $e$  edges. Let  $M$  be the maximum degree of the vertices of  $G$ , and let  $m$  be the minimum degree of the vertices of  $G$ . Show that
- a)  $2e/v \geq m$ .      b)  $2e/v \leq M$ .
- A simple graph is called **regular** if every vertex of this graph has the same degree. A regular graph is called  **$n$ -regular** if every vertex in this graph has degree  $n$ .
55. For which values of  $n$  are these graphs regular?
- a)  $K_n$       b)  $C_n$       c)  $W_n$       d)  $Q_n$
56. For which values of  $m$  and  $n$  is  $K_{m,n}$  regular?
57. How many vertices does a regular graph of degree four with 10 edges have?

In Exercises 58–60 find the union of the given pair of simple graphs. (Assume edges with the same endpoints are the same.)

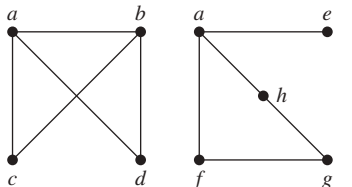
58.



59.



60.



61. The **complementary graph**  $\overline{G}$  of a simple graph  $G$  has the same vertices as  $G$ . Two vertices are adjacent in  $\overline{G}$  if and only if they are not adjacent in  $G$ . Describe each of these graphs.

a)  $\overline{K_n}$       b)  $\overline{K_{m,n}}$       c)  $\overline{C_n}$       d)  $\overline{Q_n}$

62. If  $G$  is a simple graph with 15 edges and  $\overline{G}$  has 13 edges, how many vertices does  $G$  have?
63. If the simple graph  $G$  has  $v$  vertices and  $e$  edges, how many edges does  $\overline{G}$  have?
64. If the degree sequence of the simple graph  $G$  is 4, 3, 3, 2, 2, what is the degree sequence of  $\overline{G}$ ?
65. If the degree sequence of the simple graph  $G$  is  $d_1, d_2, \dots, d_n$ , what is the degree sequence of  $\overline{G}$ ?
- \*66. Show that if  $G$  is a bipartite simple graph with  $v$  vertices and  $e$  edges, then  $e \leq v^2/4$ .
67. Show that if  $G$  is a simple graph with  $n$  vertices, then the union of  $G$  and  $\overline{G}$  is  $K_n$ .

- \*68. Describe an algorithm to decide whether a graph is bipartite based on the fact that a graph is bipartite if and only if it is possible to color its vertices two different colors so that no two vertices of the same color are adjacent.

The **converse** of a directed graph  $G = (V, E)$ , denoted by  $G^{conv}$ , is the directed graph  $(V, F)$ , where the set  $F$  of edges of  $G^{conv}$  is obtained by reversing the direction of each edge in  $E$ .

69. Draw the converse of each of the graphs in Exercises 7–9 in Section 10.1.
70. Show that  $(G^{conv})^{conv} = G$  whenever  $G$  is a directed graph.
71. Show that the graph  $G$  is its own converse if and only if the relation associated with  $G$  (see Section 9.3) is symmetric.
72. Show that if a bipartite graph  $G = (V, E)$  is  $n$ -regular for some positive integer  $n$  (see the preamble to Exercise 55) and  $(V_1, V_2)$  is a bipartition of  $V$ , then  $|V_1| = |V_2|$ . That is, show that the two sets in a bipartition of the vertex set of an  $n$ -regular graph must contain the same number of vertices.
73. Draw the mesh network for interconnecting nine parallel processors.
74. In a variant of a mesh network for interconnecting  $n = m^2$  processors, processor  $P(i, j)$  is connected to the four processors  $P((i \pm 1) \bmod m, j)$  and  $P(i, (j \pm 1) \bmod m)$ , so that connections wrap around the edges of the mesh. Draw this variant of the mesh network for 16 processors.
75. Show that every pair of processors in a mesh network of  $n = m^2$  processors can communicate using  $O(\sqrt{n}) = O(m)$  hops between directly connected processors.